

Insert new Annex 178A and Annex 178B corresponding to Clause 178 as follows:

Annex 178A

(normative)

Specification methods for 200 Gb/s per lane electrical channels

178A.1 Channel Operating Margin

178A.1.1 Introduction

The Channel Operating Margin (COM) is a figure of merit for a channel derived from a measurement of its scattering parameters (S-parameters). This annex defines a variant of the calculation defined in 93A.1 to be used by the Physical Layer and interface specifications listed in Table 178A–1. This variant of the calculation includes a different reference receiver and an associated method for parameter optimization. It also enhances the device package and termination models.

Table 178A–1—Physical Layer and interface specifications that employ Annex 178A COM

Physical Layer or interface	Reference	Parameter values
200GAUI-1 C2C 400GAUI-2 C2C 800GAUI-4 C2C 1.6TAUI-8 C2C	Annex 176C	Table 176C–10
200GAUI-1 C2M 400GAUI-2 C2M 800GAUI-4 C2M 1.6TAUI-8 C2M	Annex 176D	Table 176D–7
200GBASE-KR1 400GBASE-KR2 800GBASE-KR4 1.6TBASE-KR8	Clause 178	Table 178–15
200GBASE-CR1 400GBASE-CR2 800GBASE-CR4 1.6TBASE-CR8	Clause 179	Table 179–24

178A.1.2 Channel

The channel is illustrated in Figure 178A–1. It is composed of K signal paths indexed from 0 to $K - 1$ where $k = 0$ corresponds to the victim signal path and other indices correspond to crosstalk paths. The number of

crosstalk aggressors is a function of the structure of the channel. All significant contributors to the channel crosstalk should be included in the calculation of COM.

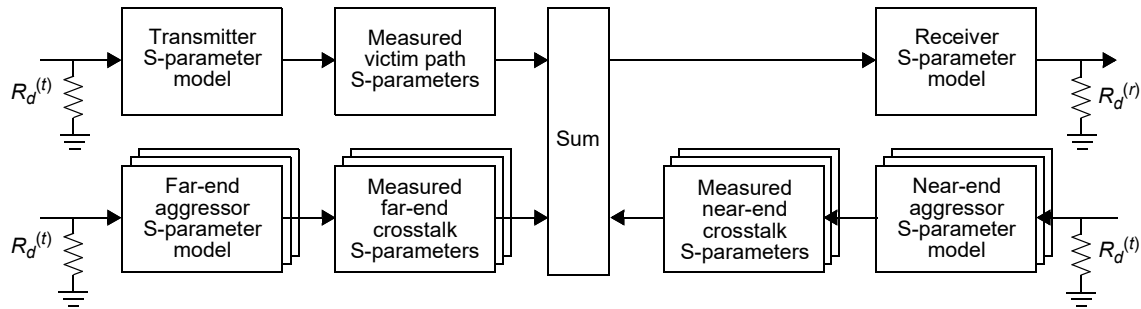


Figure 178A-1—Channel considered for the calculation of COM

Each signal path in the channel is measured according to the guidance given in 178A.1.3 to obtain the differential-mode scattering matrix defined by Equation (178A-1) at each measurement frequency f .

$$S(f) = \begin{bmatrix} s_{11}(f) & s_{12}(f) \\ s_{21}(f) & s_{22}(f) \end{bmatrix} \quad (178A-1)$$

The channel is then extended with the addition of the transmitter S-parameter model defined in 178A.1.4 and the receiver S-parameter model defined in 178A.1.5. The transmitter S-parameter model may vary between the victim transmitter, far-end aggressor transmitter, and near-end aggressor transmitter as defined by the clause or annex that specifies the use of this calculation. The S-parameters of the extended channel for signal path k , $S^{(k)}$, are defined by Equation (178A-2).

$$S^{(k)} = \text{cascade}(S^{(t)}, S, S^{(r)}) \quad (178A-2)$$

where

- $\text{cascade}()$ is the cascade connection of two-port networks defined in 93A.1.2.1
- $S^{(t)}$ is the transmitter S-parameter model
- $S^{(r)}$ is the receiver S-parameter model

The voltage transfer function for signal path k is computed from the S-parameters of the extended channel and the parameters summarized in Table 178A-2 using Equation (178A-3).

Table 178A-2—Summary of voltage transfer function parameters

Parameter	Symbol	Units
Single-ended reference resistance	R_0	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	Ω

$$H_{21}^{(k)}(f) = \frac{s_{21}^{(k)}(f)(1 - \Gamma_1)(1 + \Gamma_2)}{1 - s_{11}^{(k)}(f)\Gamma_1 - s_{22}^{(k)}(f)\Gamma_2 + \Gamma_1\Gamma_2\Delta S^{(k)}(f)} \quad (178A-3)$$

where

Γ_1 is the transmitter reflection coefficient at port 1 $(R_d^{(t)} - R_0)/(R_d^{(t)} + R_0)$
 Γ_2 is the receiver reflection coefficient at port 2 $(R_d^{(r)} - R_0)/(R_d^{(r)} + R_0)$
 $\Delta S^{(k)}(f)$ is the determinant of the S-parameter matrix $s_{11}^{(k)}(f)s_{22}^{(k)}(f) - s_{12}^{(k)}(f)s_{21}^{(k)}(f)$

178A.1.3 Measurement of the channel under test

The S-parameters for each signal path are measured between the test points specified by the clause or annex that utilizes this calculation. It is recommended that the S-parameters be measured with a uniform frequency step from a start frequency no greater than 10 MHz to a stop frequency of at least 67 GHz. The measurement frequency step corresponds to the time span of the pulse response derived from the S-parameters (see 178A.1.6). The frequency step should be chosen to be small enough so that all significant components of the pulse response are included.

Figure 178A-2 illustrates 4-port single-ended S-parameters $S^{(s)}$ and the corresponding mixed-mode S-parameters $S^{(m)}$ at any given frequency. COM is computed using the differential-mode S-parameters which are taken from the upper-left quadrant of mixed-mode S-parameter matrix. Other mixed mode S-parameters may be used as the basis for other specifications.

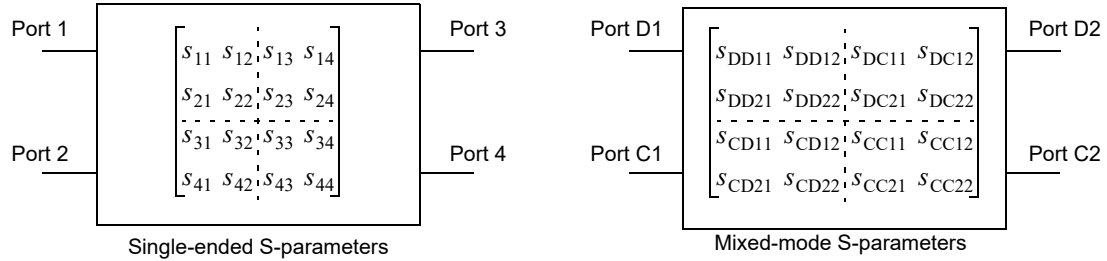


Figure 178A-2—Single-ended and mixed-mode S-parameters

The reference impedance for the single-ended S-parameters is required to be the single-ended reference resistance R_0 specified for the calculation of COM. Note that the differential reference impedance is $2R_0$ and the common-mode reference impedance is $R_0/2$. When the single-ended reference impedance for the measurement R_x differs from R_0 , the measured single-ended S-parameters $S^{(x)}$ are transformed using Equation (178A-4) where n is the number of ports represented by $S^{(x)}$.

$$S^{(s)} = A^{-1}(S^{(x)} - \rho)(I - \rho S^{(x)})^{-1}A \quad (178A-4)$$

where

A is an $n \times n$ diagonal matrix with diagonal values $\sqrt{R_0/R_x}/(R_0 + R_x)$
 I is an $n \times n$ identity matrix
 ρ is an $n \times n$ diagonal matrix with diagonal values $(R_0 - R_x)/(R_0 + R_x)$

Single-ended S-parameters are then transformed to mixed-mode S-parameters using Equation (178A-5).

$$S^{(m)} = MS^{(s)}M^{-1} \quad (178A-5)$$

where M is defined by Equation (178A-6).

$$M = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad (178A-6)$$

178A.1.4 Transmitter S-parameter model

The transmitter S-parameter model $S^{(t)}$ is the cascade connection of a device termination model, a device package model, and an optional partial host channel model as shown in Figure 178A-3. It is computed using Equation (178A-7).

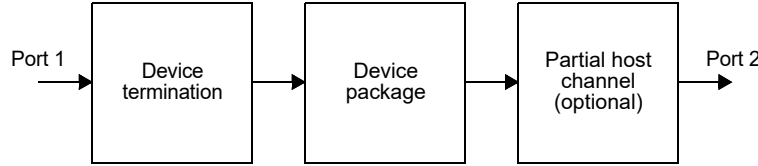


Figure 178A-3—Transmitter S-parameter model

$$S^{(t)} = \text{cascade}(S^{(d)}, S^{(p)}) \quad (178A-7)$$

where

$S^{(d)}$ is the device termination model defined in 178A.1.4.1

$S^{(p)}$ is the device package model defined in 178A.1.4.2

The partial host channel model may be included to account for the portions of a host that are not included in the channel S-parameter measurement. When the optional partial host channel model is included, $S^{(t)}$ is modified by the assignment defined by Equation (178A-8).

$$S^{(t)} = \text{cascade}(S^{(t)}, S^{(h)}) \quad (178A-8)$$

where $S^{(h)}$ is the partial host channel model defined in 178A.1.4.3.

178A.1.4.1 Device termination

The device termination model is shown in Figure 178A–4 and is a function of the parameters summarized in Table 178A–3. It is the cascade connection of N stages and a shunt capacitor C_b . Each stage consists of a shunt capacitor and series inductor.

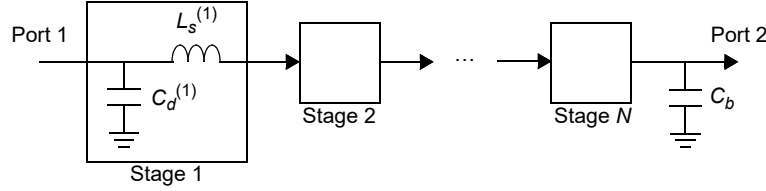


Figure 178A–4—Device termination model

Table 178A–3—Summary of device termination parameters

Parameter	Symbol	Units
Single-ended series inductance for stage n	$L_s^{(n)}$	nH
Single-ended shunt capacitance for stage n	$C_d^{(n)}$	nF
Single-ended bump capacitance	C_b	nF

The S-parameters for each stage $S^{(n)}$ are computed using Equation (178A–9).

$$S^{(n)} = \text{cascade}(S(C_d^{(n)}), S(L_s^{(n)})) \quad (178A-9)$$

where

$S(C_d^{(n)})$ are the S-parameters for shunt capacitor $C_d^{(n)}$ defined in 93A.1.2.2
 $S(L_s^{(n)})$ are the S-parameters for series inductor $L_s^{(n)}$ defined in 93A.1.2.2a

The S-parameters for the device termination model $S^{(d)}$ are initialized to $S(C_b)$. The model for each stage n is then added to the device termination model, in reverse order from $n = N$ to $n = 1$, using the assignment defined by Equation (178A–10).

$$S^{(d)} = \text{cascade}(S^{(n)}, S^{(d)}) \quad (178A-10)$$

178A.1.4.2 Device package

The device package model is shown in Figure 178A–5 and is a function of the parameters summarized in Table 178A–4. It is the cascade connection of N stages and a shunt capacitor C_p . Each stage is a model of a transmission line.

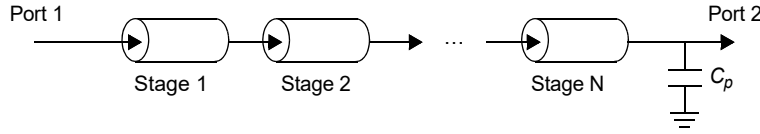


Figure 178A–5—Device package model

Table 178A–4—Summary of device package parameters

Parameter	Symbol	Units
Transmission line parameter, γ_0 , for stage n	$\gamma_0^{(n)}$	1/mm
Transmission line parameter, a_1 , for stage n	$a_1^{(n)}$	ns ^{1/2} /mm
Transmission line parameter, a_2 , for stage n	$a_2^{(n)}$	ns/mm
Transmission line parameter, τ , for stage n	$\tau^{(n)}$	ns/mm
Transmission line differential characteristic impedance for stage n	$Z_c^{(n)}$	Ω
Transmission line length for stage n	$z_p^{(n)}$	mm
Single-ended package capacitance at the package-to-board interface	C_p	nF

The S-parameters for the device package model $S^{(p)}$ are initialized to $S(C_p)$. The stage models $S^{(n)}$ are then added to the device package model, in reverse order from $n = N$ to $n = 1$, using the assignment defined by Equation (178A–11).

$$S^{(p)} = \text{cascade}(S^{(n)}, S^{(p)}) \quad (178A-11)$$

where $S^{(n)}$ are the S-parameters for transmission line stage n defined in 93A.1.2.4.

178A.1.4.3 Partial host channel

The partial host channel model is shown in Figure 178A–6 and is a function of the parameters summarized in Table 178A–5. It is the cascade connection of a first shunt capacitor C_0 , a transmission line, and a second shunt capacitor C_1 .

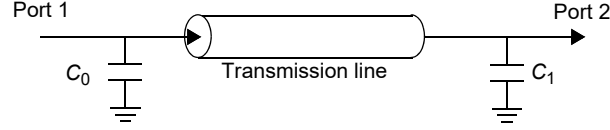


Figure 178A–6—Partial host channel model

Table 178A–5—Summary of partial host channel parameters

Parameter	Symbol	Units
Single-ended board capacitance at the package-to-board interface	C_0	nF
Host transmission line parameter, γ_0	$\gamma_0^{(h)}$	1/mm
Host transmission line parameter, a_1	$a_1^{(h)}$	ns ^{1/2} /mm
Host transmission line parameter, a_2	$a_2^{(h)}$	ns/mm
Host transmission line parameter, τ	$\tau^{(h)}$	ns/mm
Host transmission line differential characteristic impedance	$Z_c^{(h)}$	Ω
Host transmission line length	$z_p^{(h)}$	mm
Single-ended capacitance at the model-to-measurement interface	C_1	nF

The S-parameters for the partial host channel model $S^{(h)}$ are initialized to the S-parameters for a transmission line defined in 93A.1.2.4. They are then augmented by the assignment defined by Equation (178A–12).

$$S^{(h)} = \text{cascade}(S(C_0), S^{(h)}, S(C_1)) \quad (178A-12)$$

where

$S(C_0)$ are the S-parameters for shunt capacitor C_0 defined in 93A.1.2.2
 $S(C_1)$ are the S-parameters for shunt capacitor C_1 defined in 93A.1.2.2

178A.1.5 Receiver S-parameter model

The receiver S-parameter model $S^{(r)}$ is the cascade connection of an optional partial host channel model, a device package model, and a device termination model as shown in Figure 178A–7. It is computed using the procedure defined in 178A.1.4 but with the parameter values specified for the receiver model. The port order of the resulting model is then reversed so that port 1 becomes the input to the optional partial host channel

(or the device package when the partial host channel is not included) and port 2 becomes the output of the device termination.

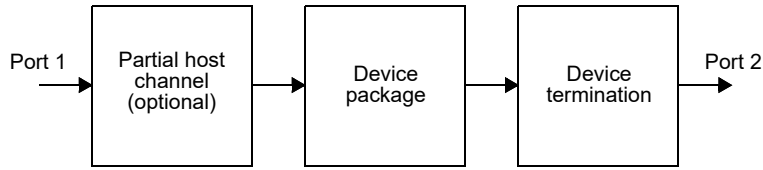


Figure 178A-7—Receiver S-parameter model

178A.1.6 Time-domain responses

For each signal path k , the voltage transfer function computed in 178A.1.2 is subject to additional filtering as defined by Equation (178A-13). The parameters for the various filter transfer functions are summarized in Table 178A-6.

$$H^{(k)}(f) = H_{txffe}(f)H_t(f)H_{21}^{(k)}(f)H_r(f)H_{ctf}(f) \quad (178A-13)$$

where

$H_{txffe}(f)$	is transmitter equalizer transfer function defined in 178A.1.6.1
$H_t(f)$	is the input rise time transfer function defined in 178A.1.6.2
$H_{21}^{(k)}(f)$	is the voltage transfer function for signal path k defined in 178A.1.2
$H_r(f)$	is the receiver noise filter transfer function defined in 178A.1.6.3
$H_{ctf}(f)$	is the receiver continuous-time equalizer transfer function defined in 178A.1.6.4

Table 178A-6—Summary of filter parameters

Parameter	Symbol	Units
Signaling rate	f_b	GBd
Differential peak output voltage, victim transmitter	A_v	V
Differential peak output voltage, far-end aggressor	A_{fe}	V
Differential peak output voltage, near-end aggressor	A_{ne}	V
Input rise time	T_r	ns
Receiver 3 dB bandwidth	f_r	GHz

The filtered voltage transfer function is then converted to a time-domain pulse response as defined by Equation (178A-14).

$$h^{(k)}(t) = \text{IDFT}[A_r T_b \text{sinc}(f T_b) H^{(k)}(f)] \quad (178A-14)$$

where

$\text{IDFT}[x]$	is the inverse discrete Fourier transform of x
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A_t	is the transmitter peak differential output voltage
T_b	is the unit interval $1/f_b$
$\text{sinc}(x)$	is the normalized sinc function of x , $\text{sinc}(x) = \sin(\pi x)/(\pi x)$

The value of A_t that is used to calculate $h^{(k)}(t)$ is a function of the path index k . For the victim signal path ($k = 0$), A_t is set to A_v . For a far-end crosstalk signal path, A_t is set to A_{fe} . For a near-end crosstalk signal path, A_t is set to A_{ne} .

NOTE—COM is calculated using measurements at discrete frequencies over a bounded frequency range (see 178A.1.3) and the inverse discrete Fourier transform is used in Equation (178A-14) to derive the corresponding discrete time signal. The filtered voltage transfer function may need to be extrapolated (both to DC and to one half of the sampling frequency) for this calculation. The time step of the discrete time signal should be no larger than $T_b/32$ (corresponding to a sampling frequency of at least $32f_b$). The extrapolation method should be chosen carefully to limit the error in the COM result.

178A.1.6.1 Transmitter equalizer

The parameters of the transmitter equalizer are summarized in Table 178A-7 and the transmitter equalizer transfer function by Equation (178A-15).

Table 178A-7—Summary of transmitter equalizer parameters

Parameter	Symbol	Units
Coefficient n	$c(n)$	
Maximum value		—
Minimum value		—
Step size		—

$$H_{txff}(f) = \sum_n c(n) e^{-j2\pi n f / f_b} \quad (178A-15)$$

The same parameter values are used to compute the transmit equalizer transfer function for the victim and far-end crosstalk signal paths. The sum of the coefficient magnitudes is 1 and combinations of coefficient values that do not satisfy this constraint are not used. The transmit equalizer transfer function for near-end crosstalk signal paths is set to 1 (no equalization).

178A.1.6.2 Input rise time filter

The input rise time filter $H_r(f)$ corresponding to a rise time T_r is defined by Equation (93A-46).

178A.1.6.3 Receiver noise filter

The receiver noise filter $H_r(f)$ with 3 dB bandwidth f_r is defined by Equation (93A-20).

178A.1.6.4 Receiver continuous-time equalizer

The parameters of the receiver continuous-time equalizer are summarized in Table 178A-8 and the equalizer transfer function is defined by Equation (178A-16).

Table 178A-8—Summary of receiver continuous-time equalizer parameters

Parameter	Symbol	Units
Pole 1 frequency	f_{p1}	GHz
Pole 2 frequency	f_{p2}	GHz
Pole 3 frequency	f_{p3}	GHz
Zero 1 frequency for $g_1 = 0$	f_{z1}	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	GHz
Gain 1 Maximum value Minimum value Step size	g_1	dB dB dB
Gain 2 Maximum value Minimum value Step size	g_2	dB dB dB

$$H_{ctf}(f) = \frac{\left(10^{g_1/20} + j\frac{f}{f_{z1}}\right)\left(10^{g_2/20} + j\frac{f}{f_{z2}}\right)}{\left(1 + j\frac{f}{f_{p1}}\right)\left(1 + j\frac{f}{f_{p2}}\right)\left(1 + j\frac{f}{f_{p3}}\right)} \quad (178A-16)$$

178A.1.7 Noise autocorrelation function

The sources of noise considered in the calculation of COM are illustrated in Figure 178A-8. Each source of noise is represented by its power spectral density at the output of the quantizer. The quantizer samples the signal and noise once per unit interval and quantizes the amplitude to discrete levels. The parameters used to compute these power spectral densities are summarized in Table 178A-9.

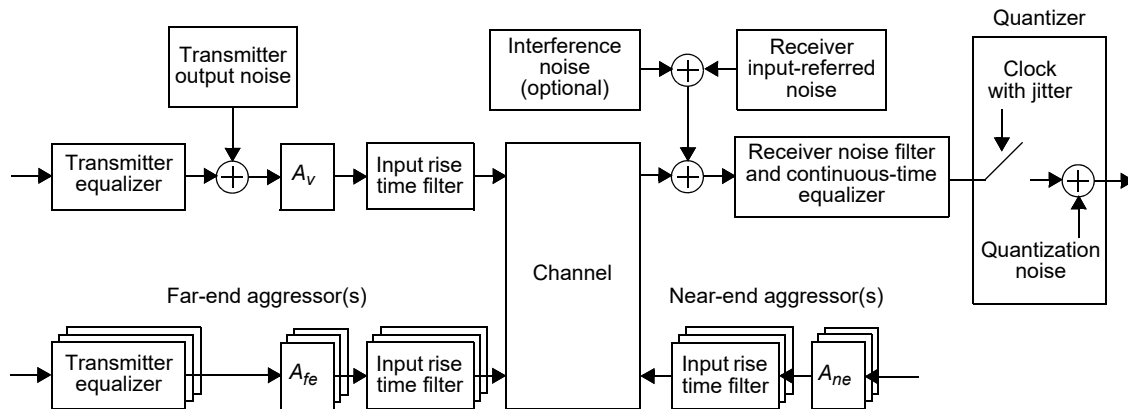


Figure 178A-8—Sources of noise considered in calculation of COM

Table 178A–9—Summary of noise parameters

Parameter	Symbol	Units
Number of signal levels	L	—
One-sided noise spectral density at receiver input	η_0	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	dB
Random jitter, RMS	σ_{RJ}	UI
Dual-Dirac jitter, peak	A_{DD}	UI
Noise-equivalent quantization bits	N_{qb}	—
Quantization clipping probability	P_{qc}	—

The total noise power spectral density S_n at the quantizer output is the sum of the individual noise power spectral densities as defined in Equation (178A–17).

$$S_n(\theta) = S_{rn}(\theta) + \sum_{k=1}^{K-1} S_{xn}^{(k)}(\theta) + S_{in}(\theta) + S_{jn}(\theta) + S_{in}(\theta) + S_{qn}(\theta) \quad (178A-17)$$

where

θ	is the normalized frequency $2\pi f/f_b$ with range $[-\pi, \pi)$
$S_{rn}(\theta)$	is the receiver input-referred noise power spectral density defined in 178A.1.7.1
$S_{xn}^{(k)}(\theta)$	is the crosstalk power spectral density for signal path k defined in 178A.1.7.2
$S_{in}(\theta)$	is the transmitter output noise power spectral density defined in 178A.1.7.3
$S_{jn}(\theta)$	is the power spectral density of the noise due to jitter defined in 178A.1.7.4
$S_{in}(\theta)$	is the interference noise power spectral density defined in 178A.1.7.5
$S_{qn}(\theta)$	is the quantization noise power spectral density defined in 178A.1.7.6

The noise autocorrelation is derived from the total noise power spectral density using Equation (178A–18).

$$R_n(n) = \text{IDFT}[S_n(\theta)]f_b \quad (178A-18)$$

178A.1.7.1 Receiver input noise

The power spectral density of the sampled receiver noise at the input to the discrete-time receiver equalizer is defined by Equation (178A–19).

$$S_{rn}(\theta) = \frac{\eta_0}{2} \sum_{m \text{ integer}} \left| H_{rn} \left(\frac{\theta + 2\pi m}{2\pi T_b} \right) \right|^2 \quad (178A-19)$$

where $H_{rn}(f)$ is $H_r(f)H_{ctf}(f)$.

178A.1.7.2 Crosstalk

The power spectral density of the sampled crosstalk signal corresponding to signal path k ($k > 0$) at the input to the discrete-time receiver equalizer is defined by Equation (178A–20).

$$S_{xn}^{(k)}(\theta) = \sigma_X^2 |\text{DFT}[h_{xn}^{(k)}(n)]|^2 / f_b \quad (178A-20)$$

where

DFT[x] is the discrete Fourier transform of x
 σ_X^2 is the power of the PAM- L signal $(L^2 - 1)/[3(L - 1)^2]$
 $h_{xn}^{(k)}(n)$ is $h^{(k)}(t_s^{(k)} + nT_b)$ where $t_s^{(k)}$ is chosen to maximize the sum of $[h_{xn}^{(k)}(n)]^2$ over n

178A.1.7.3 Transmitter output noise

For a given sampling time $t_s^{(0)}$, the power spectral density of the sampled transmitter noise at the input to the receiver discrete-time equalizer is defined by Equation (178A–21).

$$S_{in}(\theta) = 10^{-SNR_{TX}/10} |\text{DFT}[h_{in}(n)]|^2 / f_b \quad (178A-21)$$

where $h_{in}(n)$ are the sampled values $h_{noise}(t_s^{(0)} + nT_b)$ and the pulse response $h_{noise}(t)$ is defined by Equation (178A–22).

$$h_{noise}(t) = \text{IDFT}[A_v T_b \text{sinc}(fT_b) H_{noise}(f)] \quad (178A-22)$$

where $H_{noise}(f) = H_t(f)H_{21}^{(0)}(f)H_r(f)H_{ctf}(f)$.

178A.1.7.4 Noise due to transmitter jitter

For a given sampling time $t_s^{(0)}$, the slope of the victim signal path pulse response $h^{(j)}(n)$ around each sampled pulse response value is defined by Equation (178A–23).

$$h^{(j)}(n) = [h^{(0)}(t_s^{(0)} + \Delta t + nT_b) - h^{(0)}(t_s^{(0)} - \Delta t + nT_b)] / (2\Delta t) \quad (178A-23)$$

An estimate of the power spectral density of the noise due to transmitter jitter at the input to the discrete time receiver equalizer is defined by Equation (178A–24).

$$S_{jn}(\theta) = \sigma_X^2 (A_{DD}^2 + \sigma_{RJ}^2) |\text{DFT}[h^{(j)}(n)]|^2 / f_b \quad (178A-24)$$

178A.1.7.5 Interference noise

COM may be used to calibrate a test signal for receiver interference tolerance testing. In these cases, the S-parameters of the noise path are measured from the output of the interference noise source to the reference point specified by the clause or annex that utilizes this calculation. The measured S-parameters $S^{(n)}$ are then augmented by the receiver S-parameter model $S^{(r)}$ as defined by Equation (178A–25) and shown in Figure 178A–9.

$$S^{(i)} = \text{cascade}(S^{(n)}, S^{(r)}) \quad (178A-25)$$

The interference noise voltage transfer function $H_{21}^{(i)}(f)$ is computed from $S^{(i)}$ using Equation (178A–3) with Γ_1 set to 0.

The power spectral density of the sampled interference noise at the input to the discrete time receiver equalizer is defined by Equation (178A–26).

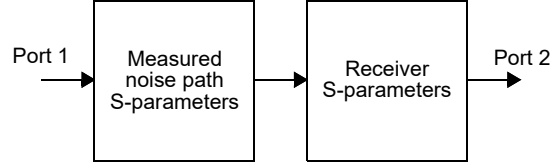


Figure 178A–9—Interference noise path S-parameter model

$$S_{in}(\theta) = \sum_{m \text{ integer}} N_{in}\left(\frac{\theta + 2\pi m}{2\pi T_b}\right) \quad (178A-26)$$

where $N_{in}(f)$ is defined by Equation (178A–27).

$$N_{in}(f) = \frac{N_s(f)}{2} |H_{21}^{(n)}(f) H_r(f) H_{ctf}(f)|^2 \quad (178A-27)$$

where $N_s(f)$ is the single-sided power spectral density at the noise generator output in V^2/Hz as defined by the clause or annex that utilizes this calculation.

When COM is not used to calibrate a test signal for receiver interference tolerance testing, $S_{in}(\theta)$ is set to 0.

178A.1.7.6 Quantization noise

The power spectral density of the quantization noise at the input of the discrete-time receiver equalizer is defined by Equation (178A–28).

$$S_{qn}(\theta) = \frac{1}{f_b} \left(\frac{\Delta^2}{12} \right) \quad (178A-28)$$

where Δ is the quantization step size defined by Equation (178A–29).

$$\Delta = 2V_{qc} / (2^{N_{qb}} - 1) \quad (178A-29)$$

where the quantization clipping amplitude V_{qc} is a positive value that satisfies the relationship $P_{sn}(-V_{qc}) = P_{qc} / 2$, $P_{sn}(y)$ is the cumulative distribution function of the noisy signal amplitude prior to quantization, and P_{qc} is the quantization clipping probability.

$P_{sn}(y)$ is derived from the probability distribution function of the noisy signal amplitude prior to quantization $p_{sn}(y)$ defined by Equation (178A–30).

$$p_{sn}(y) = \text{conv}[p_s(y), p_{ga}(y)] \quad (178A-30)$$

where $p_s(y)$ is the probability distribution function of the noiseless signal amplitude prior to quantization and $p_{ga}(y)$ is a Gaussian approximation of the probability distribution function of the noise amplitude prior to quantization.

Given a sampling time $t_s^{(0)}$, the probability distribution function of the noiseless signal amplitude prior to quantization $p_s(y)$ is computed using the procedure defined in 93A.1.7.1 substituting $h(n)$ with $h^{(0)}(t_s^{(0)} + nT_b)$ where $h^{(0)}(t)$ is defined in 178A.1.6.

The Gaussian approximation of the probability distribution function of the noise amplitude prior to quantization $p_{ga}(y)$ is defined by Equation (93A–42) substituting σ_G with σ_{ga} defined by Equation (178A–31).

$$\sigma_{ga}^2 = f_b \int_{-\pi}^{\pi} \left[S_{tn}(\theta) + \sum_{k=1}^{K-1} S_{xn}^{(k)}(\theta) + S_{tn}(\theta) + S_{jn}(\theta) + S_{in}(\theta) \right] d\theta \quad (178A–31)$$

The cumulative distribution function of the noisy signal amplitude $P_{sn}(y)$ is then computed using Equation (178A–40) substituting $p(u)$ with $p_{sn}(u)$.

178A.1.8 Optimization of variable parameters

Some parameters of the COM calculation are not assigned specific values but are instead defined by a range and a step size. These parameters are associated with the transmitter equalizer defined in 178A.1.6.1 and the receiver continuous-time equalizer defined in 178A.1.6.4. For each such parameter, a value within the specified range is chosen and the time-domain responses are computed as defined in 178A.1.6. The sampling time $t_s^{(0)}$ is then chosen and the noise autocorrelation function is computed as defined in 178A.1.7. The results of these calculations are then used to determine the values of the receiver discrete-time equalizer coefficients as defined in 178A.1.8.1 and calculate the mean-squared error as defined in 178A.1.8.2. The values of the variable equalizer parameters, and the sampling time $t_s^{(0)}$, that are used to compute COM are the values that minimize the mean-squared error.

178A.1.8.1 Receiver discrete-time equalizer coefficients

The receiver discrete-time equalizer is illustrated in Figure 178A–10. It is composed of a feed-forward filter and a feedback filter. The input to the feed-forward filter is the output of the quantizer (see Figure 178A–8). The input to the feedback filter is an estimate of the transmitted PAM- L symbol which is assumed to be

correct. The impact of an incorrect symbol estimate at the input to the feedback filter is considered in the specified target symbol error ratio.

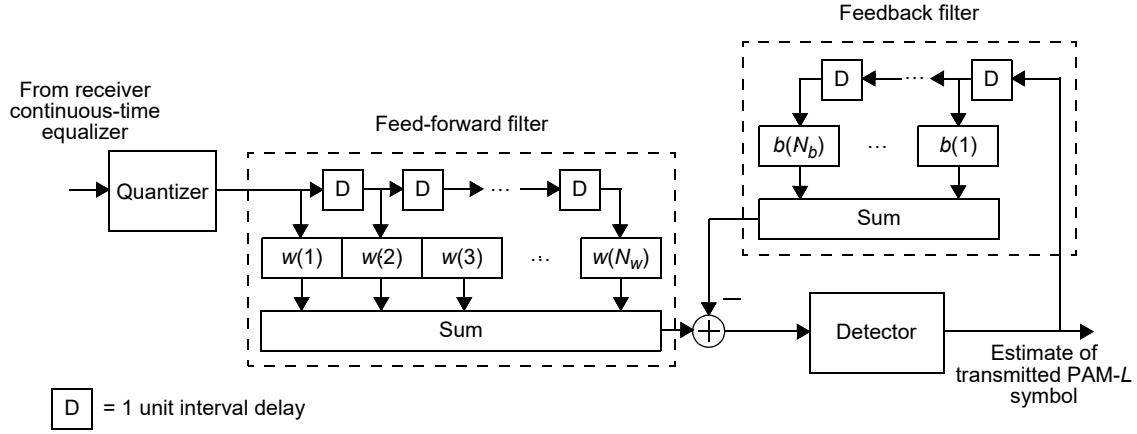


Figure 178A–10—Receiver discrete-time equalizer

The parameters of these filters are summarized in Table 178A–10. The coefficients of the filters are chosen to minimize the mean-squared error using the algorithm described below.

Table 178A–10—Summary of receiver discrete-time equalizer parameters

Parameter	Symbol	Units
Number of pre-cursor taps	d_w	—
Number of fixed-position taps	N_{fix}	—
Number of floating tap groups	N_{wg}	—
Number of taps per floating tap group	N_{wf}	—
Highest allowed tap index	N_{max}	—
Normalized upper limit on feed-forward coefficient $w(j)$	$w_{max}(j)$	—
Normalized lower limit on feed-forward coefficient $w(j)$	$w_{min}(j)$	—
Number of feedback taps	N_b	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	—

Given the pulse response of the victim signal path $h^{(0)}(t)$ and the sampling time $t_s^{(0)}$, a $1 \times N$ vector of sampled pulse responses values is defined by Equation (178A–32).

$$h = [h(-d_h) \dots h(0) \dots h(N - d_h - 1)] \quad (178A-32)$$

where $h(n)$ is $h^{(0)}(t_s^{(0)} + nT_b)$.

Define $H^{(all)}$ to be an $(N + N_{\max} - 1) \times N_{\max}$ diagonal-constant (Toeplitz) matrix whose first column is h followed by $N_{\max} - 1$ zeros and whose first row is $h(-d_h)$ followed by $N_{\max} - 1$ zeros. When N_{wg} is not defined or set to 0, then $N_{\max}$ is N_{fix} . Define $R_{nn}^{(all)}$ to be an $N_{\max} \times N_{\max}$ Toeplitz matrix whose first row and first column are $R_n(n)$ for $n = 0$ to $N_{\max} - 1$.

The total number of taps in the feed-forward filter, N_w , is $N_{fix} + N_{wg}N_{wf}$. Define i to be a $1 \times N_w$ vector consisting of indices 1 to N_{fix} for fixed-position taps and $N_{wg}N_{wf}$ additional indices for floating taps. Feed-forward filter tap index 1 corresponds to the earliest pre-cursor tap, tap index N_{fix} corresponds to the latest fixed-position post cursor tap, and $N_{\max}$ corresponds to the highest index that can be chosen for a floating tap. This means that the feed-forward filter includes $N_{fix} - d_w - 1$ fixed-position post-cursor taps and $N_{\max} - d_w - 1$ is the latest post-cursor position that can be chosen for a floating tap. Floating tap indices are chosen to minimize the mean-squared error (see 178A.1.8.2) with the constraint that indices within a floating tap group are contiguous and do not overlap with indices of fixed-position taps or other floating tap groups.

Let H be an $(N + N_{\max} - 1) \times N_w$ matrix formed from the columns of $H^{(all)}$ indexed by i . Let R_{nn} be the $N_w \times N_w$ matrix formed from the rows and columns of $R_{nn}^{(all)}$ indexed by i . Let d be $d_h + d_w$. The feed-forward and feedback filter coefficients that minimize the means-squared are determined by Equation (178A-33).

$$\begin{bmatrix} w \\ b \\ \lambda \end{bmatrix} = \begin{bmatrix} R & -H_b^T & -h_0^T \\ -H_b & I_b & z_b^T \\ h_0 & z_b & 0 \end{bmatrix}^{-1} \begin{bmatrix} h_0^T \\ z_b^T \\ 1 \end{bmatrix} \quad (178A-33)$$

where

R	is the $N_w \times N_w$ matrix $H^T H + R_{nn} / \sigma_X^2$ and H^T is the matrix transpose of H
h_0	is row $d + 1$ from H
H_b	is the $N_b \times N_w$ matrix formed from rows $d + 2$ to $d + N_b + 1$ from H
I_b	is a $N_b \times N_b$ identity matrix
z_b	is a $1 \times N_b$ vector of zeros

In Equation (178A-33), the first N_w rows of the left hand side of the equation are the feed-forward filter coefficients w , the next N_b rows are the feedback filter coefficients b , and the scalar λ is ignored. The solution to Equation (178A-33) not only minimizes the mean-squared error but also ensures that the amplitude of the equalized pulse (the value at time $d + 1$) is 1.

Limits on the feedback filter coefficients b are imposed to yield b_{lim} using Equation (178A-34).

$$b_{lim}(j) = \begin{cases} b_{\max}(j) & b(j) \geq b_{\max}(j) \\ b_{\min}(j) & b(j) \leq b_{\min}(j) \\ b(j) & \text{otherwise} \end{cases} \quad (178A-34)$$

If $b_{lim}(j)$ differs from $b(j)$ for any j , then the feed-forward filter coefficients are re-computed using Equation (178A-35).

$$\begin{bmatrix} w \\ \lambda \end{bmatrix} = \begin{bmatrix} R & -h_0^T \\ h_0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} h_0^T + H_b^T b_{lim} \\ 1 \end{bmatrix} \quad (178A-35)$$

In Equation (178A–35), the first N_w rows of the left-hand side of the equation are the updated feed-forward coefficients w and the scalar λ is again ignored.

Limits on the feed-forward coefficients w are imposed to yield w_{lim} using Equation (178A–36).

$$w_{lim}(j) = \begin{cases} w(d_w + 1)w_{\max}(j) & w(j) \geq w(d_w + 1)w_{\max}(j) \\ w(d_w + 1)w_{\min}(j) & w(j) \leq w(d_w + 1)w_{\min}(j) \\ w(j) & \text{otherwise} \end{cases} \quad (178A-36)$$

If $w_{lim}(j)$ differs from $w(j)$ for any j , then w_{lim} is divided by the scalar $h_0 w_{lim}$ where h_0 is a $1 \times N_w$ vector and w_{lim} is a $N_w \times 1$ vector, to ensure that the amplitude of the equalized pulse is 1. The feedback filter coefficients b are then recomputed to be $H_b w_{lim}$ and limits are imposed on the updated b to yield b_{lim} using Equation (178A–34).

178A.1.8.2 Calculation of mean-squared error

Given the receiver discrete-time equalizer coefficients w_{lim} and b_{lim} determined in 178A.1.8.1, and the corresponding values of R , h_0 , and H_b , the mean-squared error is computed using Equation (178A–37).

$$\sigma_e^2 = \sigma_X^2(w_{lim}^T R w_{lim} + 1 + b_{lim}^T b_{lim} - 2w_{lim}^T h_0 - 2w_{lim}^T H_b^T b_{lim}) \quad (178A-37)$$

178A.1.9 Calculation of COM

COM is the ratio, expressed in decibels, of the calculated signal amplitude to the calculated noise and interference amplitude as defined by Equation (178A–38) using the parameters summarized in Table 178A–11.

$$COM = 20\log_{10}(A_s/A_{ni}) \quad (178A-38)$$

Table 178A–11—Additional parameters for the calculation of COM

Parameter	Symbol	Units
Level separation mismatch ratio	R_{LM}	—
Target detector error ratio	DER_0	—

178A.1.9.1 Signal amplitude

The minimum distance between the signal and the detector threshold is defined by Equation (178A–39).

$$A_s = R_{LM}/(L - 1) \quad (178A-39)$$

where R_{LM} is the level separation mismatch ratio.

178A.1.9.2 Noise and interference amplitude

The probability distribution function of the noise and interference amplitude $p(y)$ is calculated using the procedure defined in 178A.1.9.3. The corresponding cumulative distribution function is defined by Equation (178A-40).

$$P(y) = \int_{-\infty}^y p(u) du \quad (178A-40)$$

The noise and interference amplitude, A_{ni} , is a positive value that satisfies the relationship $P(-A_{ni}) = DER_0$ where DER_0 is the target detector error ratio.

NOTE 1—COM is expected to be computed using a quantized amplitude axis y . The amplitude step Δy introduces quantization error in the calculated distribution function that is compounded by subsequent convolutions with other quantized distribution functions. It is recommended that Δy be no larger than 0.1% of A_s .

NOTE 2—The target detector error ratio DER_0 is $SER_1 \times L / (2L - 2)$ where SER_1 is the maximum allowed probability of the initial error in a PAM- L error burst.

178A.1.9.3 Noise and interference probability distribution function

Given the transmitter equalizer, receiver continuous-time equalizer, sampling time $t_s^{(k)}$ for each signal path k , and the coefficients of the receiver discrete-time equalizer that minimize the mean-squared error, the following procedure is used to compute the probability distribution function of noise and interference amplitude $p(y)$ for use in the calculation of COM.

The sampled values of the pulse response corresponding to signal path k at the output of the feed-forward filter are defined by Equation (178A-41).

$$h_w^{(k)}(n) = \sum_{n \in i} w_{lim}(n) h^{(k)}(n - d_w - 1) \quad (178A-41)$$

where

$h^{(k)}(n)$ is $h^{(k)}(t_s^{(k)} + nT_b)$ where $h^{(k)}(t)$ is defined in 178A.1.6
 $w_{lim}(n)$ is the feed-forward filter coefficient vector computed in 178A.1.8.1
 i is the corresponding tap index vector defined in 178A.1.8.1

The notation $n \in i$ represents n assuming each value in the index vector i and hence the summation in Equation (178A-41) is over all of the values in i .

The residual intersymbol interference is computed from $h_w^{(0)}(n)$ and the feedback filter coefficients b_{lim} defined in 178A.1.8.1 using Equation (178A-42).

$$h_{ISI}(n) = \begin{cases} 0 & n = d + 1 \\ h_w^{(0)}(n) - b_{lim}(n - d - 1) & 1 \leq n - d - 1 \leq N_b \\ h_w^{(0)}(n) & \text{otherwise} \end{cases} \quad (178A-42)$$

where N_b and d are defined in 178A.1.8.1.

The probability distribution function $p_{ISF}(y)$ for the intersymbol interference is computed using the procedure defined in 93A.1.7.3 substituting $h_{ISF}(n)$ for $h(n)$. Initialize $p(y)$ to $p_{ISF}(y)$.

The sampled values of the pulse response corresponding to each crosstalk signal path k ($k > 0$) at the output of the feed-forward filter are computed using Equation (178A–41). The probability distribution function $p_w^{(k)}(y)$ of the interference from crosstalk signal path k is then computed using the procedure defined in 93A.1.7.3 substituting $h_w^{(k)}(n)$ for $h(n)$. The value of $p(y)$ is then assigned the result of $\text{conv}[p(y), p_w^{(k)}(y)]$ where $\text{conv}[u, v]$ is the convolution of u and v . This assignment is repeated for all k greater than zero to include probability distribution function of the interference from all crosstalk signals.

The slope of the victim signal path pulse response around each sampled pulse response value at the output of the feed-forward filter is computed using Equation (178A–23) substituting $h^{(J)}(n)$, as defined in 178A.1.7.4, for $h^{(k)}(n)$. Let $h_w^{(J)}(n)$ be the result of this calculation. The probability distribution function $p_w^{(DD)}(y)$ for the noise due to transmitter dual-Dirac jitter is then computed using the procedure defined in 93A.1.7.3 substituting $A_{DD}h_w^{(J)}(n)$ for $h(n)$. Assign the value of $p(y)$ to the result of $\text{conv}[p(y), p_w^{(DD)}(y)]$ in order to include the probability distribution function for the noise due to transmitter dual-Dirac jitter.

The probability distribution function for the quantization noise amplitude at the input to the discrete-time equalizer $p_\Delta(y)$ is uniform over the range $-\Delta/2$ to $\Delta/2$ and defined by Equation (178A–43).

$$p_\Delta(y) = \begin{cases} 1/\Delta & -\Delta/2 \leq y < \Delta/2 \\ 0 & \text{otherwise} \end{cases} \quad (178A-43)$$

where Δ is defined in 178A.1.7.6 with the exception that the Gaussian approximation of the probability density function of the noise amplitude $p_{ga}(y)$ is replaced with the probability density function of the noise amplitude $p_n(y)$ defined below.

The probability density function of the noise amplitude at the input to the receiver discrete-time equalizer prior to quantization $p_n(y)$ is initialized to a Dirac delta function (equal to 1 at $y = 0$ and 0 otherwise).

The probability distribution function of the interference from crosstalk signal path k at the input to the receiver discrete-time equalizer $p^{(k)}(y)$ is then computed using the procedure defined in 93A.1.7.3 substituting $h^{(k)}(t_s^{(k)} + nT_b)$ for $h(n)$ where $h^{(k)}(t)$ is defined in 178A.1.6. The value of $p_n(y)$ is iteratively assigned the result of $\text{conv}[p_n(y), p^{(k)}(y)]$ for all k greater than zero.

The probability distribution function for the noise due to transmitter dual-Dirac jitter at the input to the receiver discrete-time equalizer $p^{(DD)}(y)$ is computed using the procedure defined in 93A.1.7.3 substituting $A_{DD}h^{(J)}(n)$ for $h(n)$. Assign the value of $p_n(y)$ to the result of $\text{conv}[p_n(y), p^{(DD)}(y)]$.

The probability distribution function of the Gaussian noise amplitude at the input to the receiver discrete-time equalizer $p_{gn}(y)$ is defined by Equation (93A–42) substituting σ_G with σ_{gn} defined by Equation (178A–44).

$$\sigma_{gn}^2 = f_b \int_{-\pi}^{\pi} [S_{rn}(\theta) + S_{tn}(\theta) + S_{jn}^{(RJ)}(\theta) + S_{in}(\theta)] d\theta \quad (178A-44)$$

where $S_{rn}(\theta)$, $S_{tn}(\theta)$, and $S_{in}(\theta)$ are defined in 178A.1.7 and $S_{jn}^{(RJ)}(\theta)$ is defined by Equation (178A–45).

$$S_{jn}^{(RJ)}(\theta) = \sigma_X^2 \sigma_{RJ}^2 |\text{DFT}[h^{(J)}(n)]|^2 / f_b \quad (178A-45)$$

The value of $p_n(y)$ is assigned the result of $\text{conv}[p_n(y), p_{gn}(y)]$. This resulting $p_n(y)$ is used to compute the quantization step size Δ used in Equation (178A–43).

The probability distribution function for the quantization noise at the detector input $p_{qn}(y)$ is initialized to a Dirac delta function and the assignment defined by Equation (178A–46) is repeated for each value n in the feed-forward filter tap index vector i (i.e., over all of the feed-forward filter coefficients).

$$p_{qn}(y) = \text{conv}\left[p_{qn}(y), \frac{1}{w_{lim}(n)} p_{\Delta}\left(\frac{y}{w_{lim}(n)}\right)\right] \quad (178A–46)$$

Assign the value of $p(y)$ to the result of $\text{conv}[p(y), p_{qn}(y)]$ in order to include the probability distribution function for quantization noise.

The combined variance of the remaining noise terms at the output of the feed-forward filter is defined by Equation (178A–47).

$$\sigma_G^2 = f_b \int_{-\pi}^{\pi} [S_{in}(\theta) + S_{jn}^{(RJ)}(\theta) + S_{rn}(\theta) + S_{in}(\theta)] |H_{rxffe}(\theta)|^2 d\theta \quad (178A–47)$$

where

$S_{jn}^{(RJ)}(\theta)$ is the power spectral density of the noise due only to transmitter random jitter defined by Equation (178A–45)

$H_{rxffe}(\theta)$ is the transfer function of feed-forward filter defined by Equation (178A–48)

$$H_{rxffe}(\theta) = \sum_{n \in i} w(n) \exp[-j\theta(n - d_w - 1)] \quad (178A–48)$$

The probability distribution function $p_G(y)$ of the corresponding Gaussian noise amplitudes is defined by Equation (93A–42). Assign the value of $p(y)$ to the result of $\text{conv}[p(y), p_G(y)]$ in order to include the probability distribution function for Gaussian noise. The result is used for the calculation of COM.

178A.1.10 Maximum likelihood sequence detection

The reference receiver may be specified to include a maximum likelihood sequence detector (MLSD) instead of the decision feedback equalizer (DFE) defined in 178A.1.8.1. In this case, the receiver discrete-time equalizer is modified as shown in Figure 178A–11.

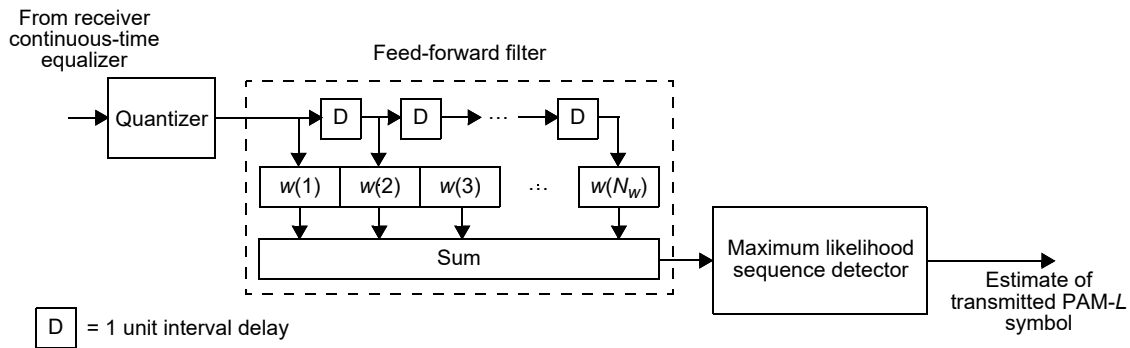


Figure 178A–11—Receiver discrete-time equalizer with MLSD

The receiver discrete-time equalizer coefficients are determined using the procedure defined in 178A.1.8.1 using the parameters defined in Table 178A–10 but with the value of N_b set to 1. COM is then computed as defined in 178A.1.9 and the resulting value is labeled COM_{DFE} . The value of COM_{DFE} and the feedback filter coefficient $b_{lim}(1)$, along with the corresponding noise and residual intersymbol interference computed at the output of the feed-forward filter, are used to calculate a modification to COM_{DFE} that represents the advantage the MLSD-based receiver has over the DFE-based receiver. This modification is defined by Equation (178A–49).

$$COM = COM_{DFE} + 20\log_{10}(-P_{an}^{-1}(DER_{MLSD})/A_s) \quad (178A-49)$$

where $P_{an}(y)$ is defined in 178A.1.10.1 and DER_{MLSD} is defined by Equation (178A–50).

$$DER_{MLSD} = \sum_{j=1}^{\infty} \left(\frac{L-1}{L} \right)^{j-1} P_j \left(-A_s \frac{(u_j^T u_j)^{3/2}}{(u_j^T V_j u_j)^{1/2}} \right) \quad (178A-50)$$

where

$P_j(y)$ is defined in 178A.1.10.2
 u_j is a $(j+1) \times 1$ vector whose elements are defined by Equation (178A–51)
 V_j is defined in 178A.1.10.3

$$u_j(i) = \begin{cases} 1 & i = 1 \\ (-1)^{i-1} (1 - b_{lim}(1)) & 2 \leq i \leq j \\ (-1)^i b_{lim}(1) & i = j + 1 \end{cases} \quad (178A-51)$$

If the value of COM calculated by Equation (178A–49) is less than COM_{DFE} , then the value of COM is set to be equal to COM_{DFE} .

Additional parameters that are used to compute COM for the MLSD-based reference receiver are listed in Table 178A–12. The “Include maximum likelihood sequence detection” parameter is set to “yes” to indicate that the calculations in this subclause are to be included in the determination of the COM value and is set to “no” otherwise.

Table 178A–12—Additional parameters for the MLSD-based reference receiver

Parameter	Symbol	Units
Include maximum likelihood sequence detection	—	—
Penalty for added receiver noise	ΔCOM_{an}	dB

178A.1.10.1 Additional receiver noise

The receiver noise is increased before COM for the MLSD reference receiver is computed. The increased noise represents additional receiver impairments allowed by the minimum COM limit but not explicitly included in the calculation of COM_{DFE} . Let $S_{an}(\theta)$ be the power spectral density of the additional noise as defined by Equation (178A–52).

$$S_{an}(\theta) = g_{an} S_{rn}(\theta) |H_{rxffe}(\theta)|^2 \quad (178A-52)$$

where g_{an} is a scale factor for the added noise with a value greater than or equal to 0.

Let $p_{an}(y)$ be the probability distribution function of noise and interference amplitude that includes the added noise. It is computed as defined in 178A.1.9.2, using the same parameter values that were used to compute COM_{DFE} , but replacing σ_G with σ_{an} where σ_{an} is defined by Equation (178A-53).

$$\sigma_{an}^2 = \sigma_G^2 + f_b \int_{-\pi}^{\pi} S_{an}(\theta) d\theta \quad (178A-53)$$

The corresponding cumulative distribution function $P_{an}(y)$ is defined by Equation (178A-40) substituting $p_{an}(y)$ for $p(y)$.

The value of g_{an} that is used to calculate COM for the MLSD reference receiver is the value that makes the quantity $20\log_{10}[P_{an}^{-1}(DER_0)/P^{-1}(DER_0)]$ equal to the specified value of ΔCOM_{an} . If the value of ΔCOM_{an} is not specified, it defaults to the minimum COM limit defined by the clause or annex that specifies the use of this calculation.

178A.1.10.2 Sequence noise distribution function

The probability distribution function of the noise corresponding to a sequence of length j is denoted $p_j(y)$. It is initialized by the assignment defined in Equation (178A-54).

$$p_j(y) = \text{conv}\left[p_{an}(y), \frac{1}{|b_{lim}(1)|} p_{an}\left(\frac{y}{b_{lim}(1)}\right)\right] \quad (178A-54)$$

where $p_{an}(y)$ defined in 178A.1.10.1 and $\text{conv}[u, v]$ denotes the convolution of u and v .

It is iteratively updated by repeating the assignment defined by Equation (178A-55) $j - 1$ times.

$$p_j(y) = \text{conv}\left[p_j(y), \frac{1}{|1 - b_{lim}(1)|} p_{an}\left(\frac{y}{1 - b_{lim}(1)}\right)\right] \quad (178A-55)$$

The cumulative distribution function of the noise corresponding to a sequence of length j , $P_j(y)$, is then computed using Equation (178A-40) substituting $p_j(y)$ for $p(y)$.

NOTE—The probability distribution function $p(y/a)/|a|$, where $|a|$ denotes the absolute value of a , represents random variable Y scaled by a factor of a . The integral of the scaled probability distribution function is 1.

178A.1.10.3 Intersymbol interference and noise autocorrelation function

The power spectral density of the noise and residual intersymbol interference at the output of the feed-forward filter is defined by Equation (178A-56).

$$S_{ni}(\theta) = S_n(\theta) |H_{rxffe}(\theta)|^2 + S_{an}(\theta) + \sigma_X^2 |\text{DFT}[h_{ISI}(n)]|^2 / f_b \quad (178A-56)$$

where $S_{an}(\theta)$ is defined in 178A.1.10.1.

The corresponding autocorrelation function $R_{ni}(n)$ is computed using Equation (178A-18) substituting $S_{ni}(\theta)$ for $S_n(\theta)$. The normalized autocorrelation matrix V_j is a $(j+1) \times (j+1)$ Toeplitz matrix whose first row and column are $R_{ni}(n) / R_{ni}(0)$ for $n = 0$ to j .

178A.2 Channel signal to common-mode ratio

Channel signal to common-mode ratio, $SCMR_{CH}$, is calculated using the procedure in this subclause.

The parameters used to compute the channel signal to common-mode ratio are summarized in Table 178A–13. The values of these parameters are specified by the clause or annex that utilizes this calculation.

Table 178A–13—Summary of channel signal to common-mode ratio parameters

Parameter	Symbol	Units
Signaling rate	f_b	GBd
Input rise time	T_r	ns
Receiver 3 dB bandwidth	f_r	GHz
Probability that the common-mode voltage exceeds $V_{CM_{ch}}$	P_{peak}	—

Channel signal to common-mode ratio $SCMR_{CH}$ is defined by Equation (178A–57).

$$SCMR_{CH} = 20\log_{10}\left(\frac{\sigma_{ts}}{V_{CM_{ch}}}\right) \quad (178A-57)$$

where

σ_{ts} is the RMS value of the signal defined by Equation (178A–58).
 $V_{CM_{ch}}$ is the common-mode peak-to-peak voltage at the channel output due to channel differential-mode to common-mode conversion defined by Equation (178A–62).

$$\sigma_{ts} = \sqrt{\sum_n [h_{DD21}(t_{max} + nT_b)]^2} \quad (178A-58)$$

where t_{max} is the value of t that maximizes $h_{DD21}(t)$ and $h_{DD21}(t)$ is defined by Equation (178A–59).

$$h_{DD21}(t) = \text{IDFT}[T_b \text{sinc}(fT_b)H_i(f)s_{DD21}(f)H_r(f)] \quad (178A-59)$$

where

IDFT[x] is the inverse discrete Fourier transform of x
 $H_i(f)$ is the input rise time transfer function defined in 178A.1.6.2
 $s_{DD21}(f)$ is the channel differential transmission S-parameter defined in 178A.1.3
 $H_r(f)$ is the receiver noise filter transfer function defined in 178A.1.6.3
 T_b is the unit interval $1/f_b$
 $\text{sinc}(x)$ is the normalized sinc function of x , $\text{sinc}(x) = \sin(\pi x)/(\pi x)$

Calculate the common-mode signal distribution $p(y)$ using the procedure in 93A.1.7.1, with $L = 4$ and $h(n) = h_{CD21}(n)$ where $h_{CD21}(n)$ is defined by Equation (178A–60).

$$h_{CD21}(n) = h_{CD21}(t_s + nT_b) \quad (178A-60)$$

where t_s is the value of t that maximizes the value of $\sum_n [h_{CD21}(n)]^2$ and $h_{CD21}(t)$ is defined by Equation (178A–61).

$$h_{CD21}(t) = \text{IDFT}[T_b \text{sinc}(fT_b) H_t(f) s_{CD21}(f) H_r(f)] \quad (178A-61)$$

where $s_{CD21}(f)$ is the channel differential to common-mode transmission S-parameter defined in 178A.1.3.

Calculate the cumulative distribution function $P(y)$ from $p(y)$ using Equation (178A–40). The channel common-mode noise voltage, VCM_{ch} , is a positive value that satisfies the relationship defined by Equation (178A–62).

$$P(-VCM_{ch}) = P_{\text{peak}} \quad (178A-62)$$

where P_{peak} is the probability that the common-mode voltage exceeds VCM_{ch} .